

KUMLINGE ISLAND, Finland

WMO: 029900

Lat: 60.300N

Lon: 20.750E

Elev: 16

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 86-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	1.4	7.5	-2.1	4.9	2.1	4.0	6.8	7.4	42.9	35.3	37.4	37.2	11.4	30	0.815

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	73.1	64.5	70.2	62.9	67.4	61.5	65.7	71.3	63.8	68.7	62.2	66.6	10.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.3	87.2	68.8	61.6	82.0	66.9	60.1	77.6	65.0	30.5	71.5	29.1	68.7	28.0	66.9	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 39.3	33.8	30.0	5.2	77.8	14.2	2.9	-5.1	79.9	-13.4	81.5	-21.4	83.1	-31.8	85.2		
(5)			4.3	67.9	12.9	2.2	-5.0	69.5	-12.6	70.8	-19.8	72.0	-29.2	73.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.8	28.6	26.5	30.6	37.6	46.1	55.0	60.6	59.8	53.1	45.3	37.8	31.6		
	DBStd	13.25	9.37	9.46	6.99	5.07	5.18	5.06	4.15	4.32	4.57	4.82	5.40	5.89		
	HDD50	3554	662	659	600	371	145	10	0	19	152	366	570			
	HDD65	8121	1127	1079	1065	821	586	302	149	173	358	610	816	1035		
	CDD50	933	0	0	0	0	24	160	328	302	112	7	0	0		
	CDD65	25	0	0	0	0	0	3	12	10	0	0	0	0		
	CDH74	56	0	0	0	0	0	6	34	16	0	0	0	0		
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0		
	WSAvg	14.8	18.0	15.7	14.2	13.8	12.2	11.8	12.6	13.2	14.4	17.8	16.2	17.7		
	PrecAvg	32.9	3.4	2.5	2.1	1.7	1.7	2.4	2.9	3.1	2.8	3.3	3.4	3.7		
(15) Precipitation	PrecMax	40.9	5.8	5.4	3.9	2.8	3.6	4.4	5.4	5.7	5.6	6.7	7.8	6.5		
	PrecMin	24.3	0.5	0.1	0.4	0.2	0.4	1.0	0.2	1.1	0.8	0.5	0.5	0.7		
	PrecStd	4.0	1.5	1.3	0.8	0.7	0.8	0.9	1.3	1.2	1.5	1.5	1.5	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.1	41.6	45.0	56.8	67.6	74.4	78.2	76.1	68.1	55.6	49.4	43.4		
		MCWB	38.9	39.3	40.7	49.5	58.1	64.8	67.5	66.2	61.9	53.2	47.5	41.4		
	2%	DB	39.2	39.3	41.3	52.6	61.9	70.2	73.6	72.0	64.6	53.9	47.8	41.5		
		MCWB	37.0	37.8	38.5	46.7	54.2	61.3	64.6	64.7	59.3	51.9	46.2	40.1		
	5%	DB	37.9	37.5	39.3	49.0	58.4	66.5	70.9	68.9	61.7	52.7	46.3	40.5		
		MCWB	36.3	36.3	37.3	44.6	51.1	58.9	63.6	63.3	57.6	50.9	44.8	39.3		
	10%	DB	36.8	36.1	37.6	45.5	54.8	63.3	68.0	66.4	59.3	51.4	44.8	39.1		
		MCWB	35.5	35.1	36.0	42.1	49.1	57.5	62.0	61.7	55.8	49.7	43.4	37.9		
	0.4%	WB	39.6	40.1	41.3	50.0	59.0	66.2	69.2	68.0	63.3	54.0	48.0	42.0		
		MCDB	40.5	41.2	43.5	55.9	67.1	73.5	74.7	74.0	66.9	54.8	48.9	42.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.8	38.1	39.1	47.3	54.7	62.8	65.9	65.7	60.2	52.4	46.5	40.6		
		MCDB	38.5	38.8	40.7	51.9	61.6	68.0	71.5	70.8	63.6	53.4	47.6	41.3		
	5%	WB	36.7	36.6	37.6	44.8	51.7	59.9	64.0	63.8	58.1	51.2	45.0	39.5		
		MCDB	37.5	37.3	39.1	48.8	57.2	64.7	69.7	67.8	60.9	52.4	46.0	40.3		
	10%	WB	35.6	35.3	36.2	42.3	49.5	58.0	62.3	62.3	56.4	50.0	43.6	38.1		
		MCDB	36.6	36.0	37.4	45.2	54.2	62.7	67.4	65.7	58.8	51.1	44.6	39.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	7.3	7.1	8.7	10.7	9.5	9.5	7.8	7.1	5.2	5.2	5.7		
		MCDBR	5.7	5.7	8.4	15.0	17.0	15.9	14.8	12.0	9.6	6.5	5.0	5.4		
		MCWBR	5.5	5.0	6.2	10.1	10.2	8.8	7.4	6.0	6.1	5.8	5.0	5.4		
	5% WB	MCDBR	5.3	5.3	7.6	14.7	16.2	13.9	13.3	10.6	8.7	6.4	5.1	5.5		
		MCWBR	5.4	5.0	5.9	10.2	10.1	8.5	7.2	5.8	6.2	6.0	5.4	5.6		
(40) Clear-Sky Solar Irradiance	taub	0.275	0.311	0.330	0.369	0.367	0.358	0.387	0.376	0.347	0.333	0.300	0.385			
	taud	2.175	2.248	2.300	2.324	2.421	2.490	2.435	2.451	2.514	2.411	2.217	2.639			
	Ebn at Noon	153	215	252	262	273	277	265	259	248	208	140	107			
	Edh at Noon	13	22	28	33	32	31	32	29	23	19	13	9			
(44) All-Sky Solar Radiation	RadAvg	74	265	758	1280	1755	1936	1797	1399	853	362	98	45			
	RadStd	10	47	85	121	149	129	161	121	96	54	20	6			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)	+0.67	N/A	N/A	N/A	N/A	N/A	-205	N/A	N/A	N/A

Nomenclature: See separate page